variance is explained by the asset allocation decision. Their result is stated in terms of variances:[6]

$$\underbrace{\text{var}(r)}_{100\%} = \underbrace{\text{var}(r_{active})}_{\sim 10\%} + \underbrace{\text{var}(r_{bmk})}_{\sim 90\%}, \tag{14.2}$$

where approximately 90% of the variance of total returns, var(r), is accounted for by the variance of passive or benchmark returns.

For Norway, the effect of factors is even more dramatic than the set of funds studied by Brinson, Hood, and Beebower. Figure 14.2 reports the variance attribution of Norway's fund returns into its active returns and benchmark returns for the whole fund, and the equities and fixed income portfolio components of the fund as computed in the Professors' Report. From January 1998 to September 2009, 99.1% of variation in the fund's returns was explained by the equity-bond decision taken by Parliament. For the equities and bond portfolios, the passive benchmark accounted for 99.7% and 97.1% of return variation, respectively. Norway's factor attributions are so high because NBIM worked under very strict risk constraints that allowed it only extremely small deviations from the benchmark. In effect, Norway is a huge index fund.

The asset allocation decision—the asset classes where you collect your risk premiums—is by far the most important decision in understanding the fluctuations of your fund over time. If you want to explain why your fund's performance is different relative to your competitor's, however, your fund's benchmark allocation becomes less important.[7] Many endowments, for example, have similar asset allocation policies (see chapter 1), and thus the active bets taken by Harvard are a major determinant in whether Harvard beats Yale in one particular year. But for just your fund compared to itself over time, the most important investment decision you make is the top–down choice on asset allocation.

2.3. MORE SOPHISTICATED RETURN DECOMPOSITIONS

Risk managers often use more sophisticated return decompositions by drilling down into asset-class or position-level data.[8] (These are derived by subtracting and adding one just as in equation (14.1).) In all of these decompositions, the asset class decision is the most important taken by the fund.

[6] There is a covariance term in equation (14.2) that is folded into either the active or the benchmark component. The covariance term is zero if the benchmark is risk-adjusted (see chapter 10), in which case the benchmark return represents exposure to *systematic risk* and the active return corresponds to *idiosyncratic risk*.

[7] See Hensel, Ezra, and Ilkiw (1991) and Ibbotson and Kaplan (2000). Brown, Garlappi, and Tiu (2010) report that 70% of the (time-series) variance of endowment returns is due to passive policy decisions.

[8] See also Karnosky and Singer (1994) and Lo (2008).